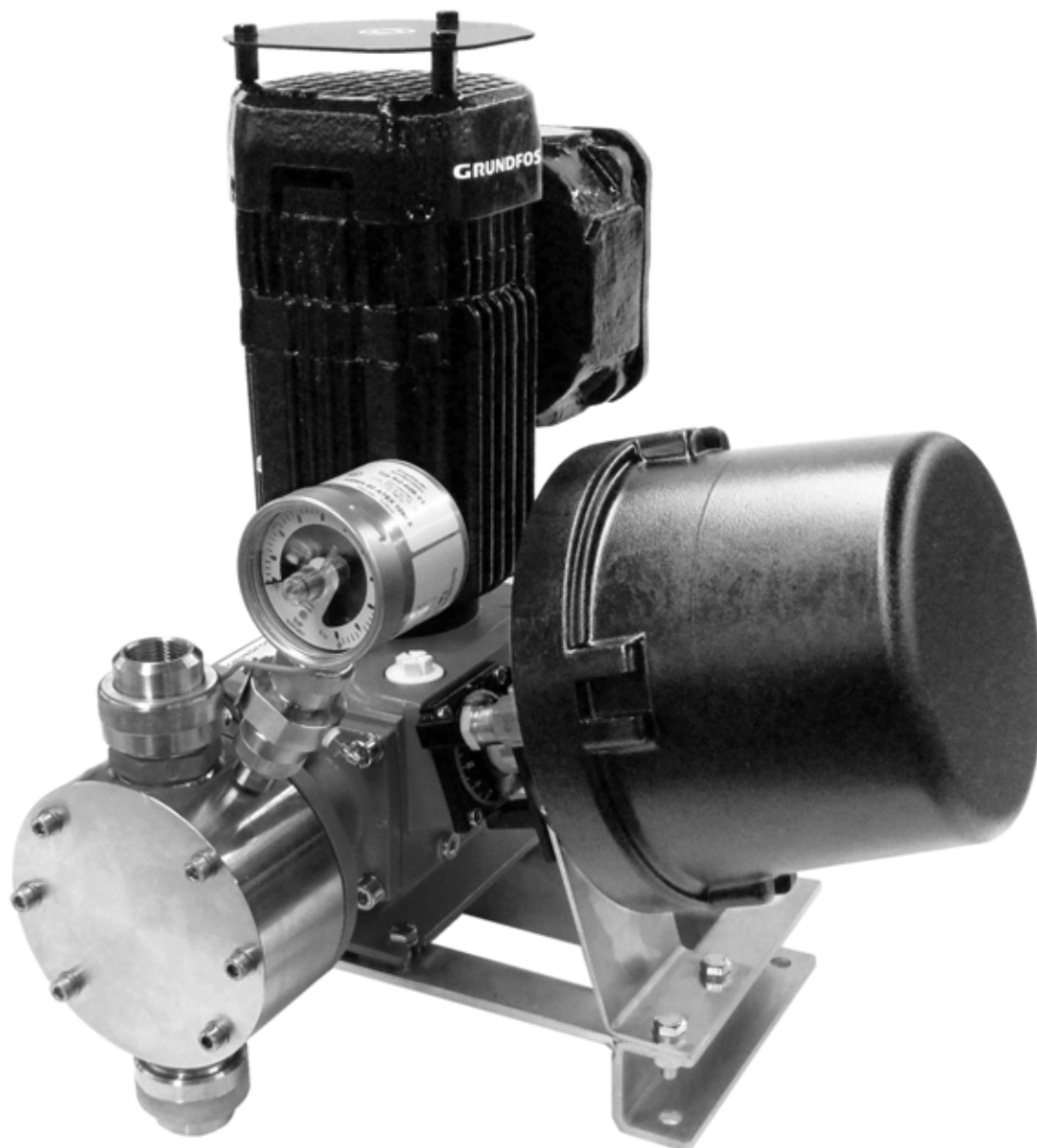


ATEX-Servomotors

with Ex protection

Installation and operating instructions



Original installation and operating instructions

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Warning

These installation and operating instructions are also available on www.grundfos.com.

Prior to installation, read these installation and operating instructions and the servomotor manufacturer's instructions. Installation and operation must comply with local regulations and accepted codes of good practice.



1. General information

These installation and operating instructions describe additional conditions or special considerations for the operation of pumps with electric servomotor for ATEX DMH pumps.

These installation and operating instructions are required in addition to the servomotor manufacturer's instructions supplied with the servomotor.

If you require further information or if any problems arise, which are not described in detail in this manual, please contact the nearest Grundfos company.

1.1 Service documentation

If you have any questions, please contact the nearest Grundfos company or service workshop.

1.2 Declaration according to ATEX GUIDELINES, 4TH EDITION - September 2012, concerning the assembling of pump and servomotor

Grundfos declares that there is no additional ignition hazard as a result of assembling the pump and servomotor. Pump assembled with servomotor is considered a combination of individual items of equipment in terms of explosion protection. Therefore, the manufacturer of the pump and the manufacturer of the electrical servomotor supply an EC declaration of conformity for each of both items.

2. Applications

Grundfos servomotors, series 384, are designed for automatic adjustment of the stroke length and consequently the dosing rate of the Grundfos dosing pumps described in this manual.

ATEX servomotors are suitable for ATEX DMH single-head pumps only.



Warning

Use the servomotor only for the specified pump types!

Caution

Other applications are not approved and not permitted. Grundfos cannot be held liable for any damage resulting from incorrect use.

3. Safety

This manual contains general instructions that must be observed during installation, operation and maintenance of the servomotor. This manual must therefore be read by the installation engineer and the relevant qualified personnel/operators prior to installation and start-up and must be available at the mounting location of the motor at all times.

For all work on the servomotor, observe the manufacturer's instructions supplied with the servomotor.

It is not only the general safety instructions given in this "Safety" section that must be observed, but also all the specific safety instructions given in the other sections.

3.1 Identification of safety instructions in this manual

If the safety instructions or other advice in this manual are not observed, it may result in personal injury or malfunction and damage to the servomotor. The safety instructions and other advice are identified by the following symbols:

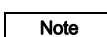


Warning

If these safety instructions are not observed, it may result in personal injury!



Caution If these safety instructions are not observed, it may result in malfunction or damage to the equipment!



Note Notes or instructions that make the job easier and ensure safe operation.

Information provided directly on the servomotor must be maintained in a readable condition at all times.

3.2 Qualification and training of personnel

The personnel responsible for the installation, operation, maintenance and inspection must be appropriately qualified for these tasks. Areas of responsibility, levels of authority and the supervision of the personnel must be precisely defined by the operations manager.

If the personnel do not have the necessary knowledge, the necessary training and instruction must be given. It is the responsibility of the operations manager to make sure that the contents of this manual are understood by the personnel.

3.3 Risks when safety instructions are not observed

Non-observance of the safety instructions may have dangerous consequences for the personnel, the environment and the pump. If the safety instructions are not observed, all rights to claims for damages may be lost.

Non-observance of the safety instructions may lead to the following hazards:

- failure of important functions of the pump/system
- failure of specified methods for maintenance
- harm to humans from exposure to electrical, mechanical and chemical influences
- damage to the environment from leakage of harmful substances.

3.4 Safety-conscious working

The safety instructions in this manual, applicable national health and safety regulations and any internal working, operating and safety instructions specified by the operations manager must be observed.

3.5 Safety instructions for the operations manager/user

Hazardous hot or cold parts on the servomotor must be protected to prevent accidental contact.

Damage caused by electrical energy must be prevented (for more details, see for example the regulations of the VDE and the local electricity supply company).

3.6 Safety instructions for installation, maintenance, and inspection work

The operations manager must ensure that all installation, maintenance and inspection work is carried out by authorised and qualified personnel, who have been adequately trained by reading this manual and the servomotor manufacturer's instructions.

All work on the servomotor should only be carried out when the servomotor is stopped.

All safety and protective equipment must be immediately restarted or put into operation once work is complete.



Warning

All work on the servomotor should be carried out in a non-explosive area or only if it can be proven that the area in which repairs are to be carried out is not a potentially explosive atmosphere.



Warning

Electrical connections must only be carried out by qualified personnel!
Repairs must only be carried out by authorised and qualified personnel!



Warning

When dosing dangerous media, observe the corresponding safety precautions!
Wear protective clothing (gloves and goggles), if necessary!
Before mounting the servomotor, switch off the pump, and disconnect it from the power supply!

3.7 Unauthorised modification and manufacture of spare parts

Modification or changes to the servomotor are only permitted following agreement with the manufacturer. Original spare parts and accessories authorised by the manufacturer are safe to use. Using other parts may result in liability for any resulting consequences.

3.8 Improper operation

The operational safety of the supplied servomotor is only ensured if it is used in accordance with section 4. *Technical data*. The specified limit values must under no circumstances be exceeded.

4. Technical data

4.1 Versions, servomotor product numbers and corresponding pump types

A servomotor kit includes servomotor, console and mounting material. A spare servomotor is delivered without console and mounting material.

Servomotor	For pump type	Pump type key, control variant	Servomotor kit	Spare servomotor	Control variant	Supply voltage
384-511	DMH 251-253, DMH 280-281	AT1	96725812	98711372	Potentiometer	115-230 V, 50/60 Hz
		AT6	96624757	98711374	Current signal	115-230 V, 50/60 Hz
384-512	DMH 254-255, DMH 283	AT1	96736527	98711372	Potentiometer	115-230 V, 50/60 Hz
		AT6	96624761	98711374	Current signal	115-230 V, 50/60 Hz
384-513	DMH 257, DMH 285-287	AT1	95702319	98711372	Potentiometer	115-230 V, 50/60 Hz
		AT6	96624765	98711374	Current signal	115-230 V, 50/60 Hz

4.2 General data

- Permissible ambient temperature: 0 °C to +60 °C
- Permissible air humidity: max. relative humidity: 70 % at 40 °C, 90 % at 35 °C.
- Turning angle: 270 °
- Actuating time: 90 s
- Enclosure class: IP65
- Ex protection:  II2G Ex d IIC T6 Gb
- With mounting console

Feedback potentiometer versions

- Single potentiometer 1000 Ω

Electronics for current signal control

- Input for 4-20 mA current signal

4.3 Electrical data

Supply voltage	Rated current [A]	Power consumption [W]
115-230 V, 50/60 Hz	0.1-0.22	25

4.4 Dimensional sketches

4.4.1 Servomotor range 384-511 to 513

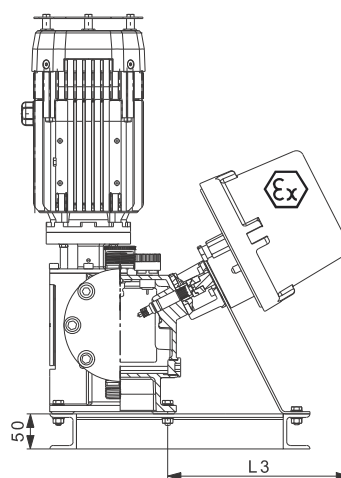


Fig. 1 Servomotor range 384-511 to 513

Servomotor range	L3
384-511, 115-230 V	297
384-512, 115-230 V	278
384-513, 115-230 V	273

Measurements in mm.

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5. Transport and storage

Do not throw or drop the servomotor.

Caution Do not use the protective packaging as transport packaging.

5.1 Unpacking

- Retain the packaging for future storage or return, or dispose of the packaging in accordance with local regulations.
- Mount the servomotor as soon as possible after unpacking.

5.2 Storage

Store the servomotor in a dry and cool place.

6. Installation

6.1 General information



Warning

The servomotor must be mounted by qualified personnel.



Warning

Tools meeting the requirements of EN 1127-1 Appendix A must be used for all operations on pumps in potentially explosive atmospheres.

There is a general ban on the use of any type of steel tool in Zone 1 if there is a risk of explosion due to materials in Explosion Group IIB and hydrogen sulphide, ethylene oxide or carbon monoxide.

The operator is responsible for setting up the system in accordance with the requirements of Directive 99/92/EC, taking into account the appropriate standards for its application.

Caution

Make sure you adhere to applicable national regulations when installing the system.



Warning

It is a basic fact that certain parts will carry a dangerous voltage whenever electrical devices are operated. Only electricians or trained persons instructed and supervised by a specialist electrician may carry out work on electrical installations or equipment and the relevant electrotechnical rules must be observed.



Warning

When undertaking installation, commissioning and testing work, all applicable safety and accident prevention regulations must always be observed.



Warning

Before you start any (installation) work of any kind on the drive, make sure that all affected machines/ installations are switched off.

Do not use the position adjustment knob at the servomotor. When using the position adjustment knob, the min./max. position can be exceeded. Exceeding the min./max. position leads to destruction of the adjusting spindle and the sliding plug in the pump.

Caution

Note

It is essential that you read this instruction manual prior to carrying out installation and commissioning.

6.2 Mounting

Note

When the pump and servomotor are ordered together, the servomotor will be supplied fitted to the pump ready for connection.

6.2.1 Retrofitting a dosing pump with an electrical servomotor

Warning



When retrofitting a dosing pump with an electrical servomotor, you must observe the instructions and warnings contained in section 6.4 *Electrical connections*, section 7. *Commissioning* and section 8. *Operation*.

- Carry out the electrical connections according to the servomotor manufacturer's instructions.

Scope of delivery of the electrical servomotor

Lantern, front plate, scale ring with screw, coupling (4 parts) with two grub screws, console, holding plate. Holding screws for console, holding plate, lantern and servomotor.

Required tools

Hexagon socket wrench 2 mm, 5 mm, wrench 13 mm

Checking the 0 % position

The servomotor is set to the 0 % position prior to delivery. However, for safety's sake this 0 % position must be checked and, if necessary, corrected.

For a servomotor with reversing potentiometer:

1. Connect the voltage supply for direction "close/min." according to the servomotor manufacturer's instructions.
2. Switch on the mains.
3. The servomotor moves to the 0 % position.
4. Disconnect the mains and remove the mains cable.

For a servomotor with analogue control:

1. Connect the voltage supply according to the servomotor manufacturer's instructions.
2. Switch on the mains.
3. Apply 4 mA to "4-20 mA input".
4. The servomotor moves to the 0 % position.
5. Disconnect the mains and remove the mains cable.

For all servomotors:

1. Connect the pump to the electrical mains.
2. Switch on the mains.
3. Turn the pump's stroke adjustment button to the 0 % position.
4. Switch off the mains.

Removing the pump's front plate

1. Remove the cap from the adjustment button.
2. Remove the holding screw and pull the adjustment button straight out.
 - Make sure that you do not alter the 0 % position.
3. Remove the front plate.

Installation sketch for the electrical servomotor

Note

Observe the installation and operating instructions for the dosing pump.

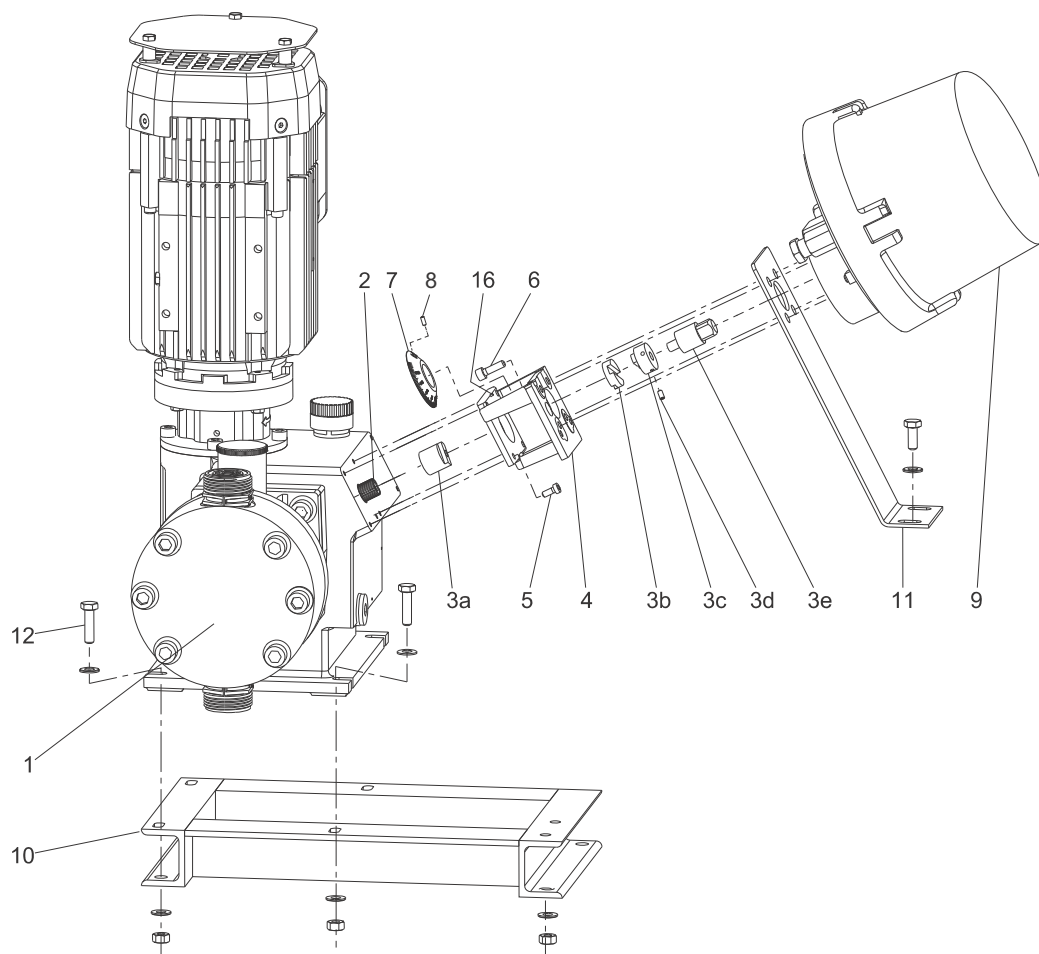


Fig. 2 Installing the servomotor

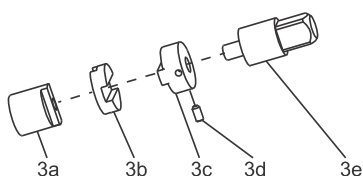


Fig. 3 Coupling

Item	Description
1	Pump
2	Adjusting spindle (grooved dowel)
3	Coupling
4	Lantern
5	Screw
6	Nut and intermediate disk
7	Scale ring
8	Grub screw
9	Servomotor
10	Console
11	Holding plate
12	Screw

Fitting the servomotor and lantern to the pump

- Place the pump (1) on the console (10) and fix the screws (12) without fastening them.
- Fit the coupling section (3a) onto the adjusting spindle (2).
- Fit the lantern (4) using four screws (5).

- Insert the scale ring (7) and coupling section (3b) (carrier) through the side of the lantern. Push one coupling section (3b) onto the other coupling section (3a) so it is in the correct position (see fig. 3) and hold it in place with one hand.

Caution The spindle must not rotate during this process.

- Turn coupling section (3c) on the shaft so that it engages in the groove on coupling section (3b).
- Fit coupling section (3e) onto coupling section (3c) and tighten slightly with the grub screw (3d).
- Screw the holding plate (11) onto the console to place it tightly at the lantern.
- Screw the servomotor (9) to the lantern and holding plate with the screws (6).
- Tighten the grub screw (3d).
- Match up the 0 % position of the scale ring (7) with the marking pin on the lantern and tighten the grub screw (8).
- Tighten the screws (12) at the pump to fasten the pump to the console.

Connecting and commissioning the servomotor

Establish the electrical connections according to the servomotor manufacturer's instructions, see also section [6.4 Electrical connections](#).

Commission the pump and servomotor, observing the information contained in the instruction manual; see also section [7. Commissioning](#).

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6.3 Earthing

Caution The servomotor must be earthed.

6.4 Electrical connections



Warning

Electrical connections may only be established by qualified staff.

Observe the servomotor manufacturer's instructions supplied with the servomotor.



Warning

Before opening housing, disconnect the servomotor from the power supply.

Observe the local safety regulations. Protect the cable connections and plugs from corrosion and moisture.



Warning

The applicable regulations must be observed when carrying out electrical installation and commissioning work. When undertaking electrical installation and commissioning work on explosion-protected equipment, the relevant national regulations on setting up electrical installations in potentially explosive areas need to be taken into account too.



Warning

Protective measures must be put in place in accordance with the relevant VDE guidelines and power supply company regulations. In particular, VDE guideline 0105, "Working on live installations", must be taken into account.



Warning

All elements, such as switch cams, switches, potentiometers, relays, etc., are wired at the factory. The internal wiring must never be modified.



Warning

The terminals at the limit switches are live when the switch closes.



Warning

The following must be taken into account for explosion-protection servomotors and pumps.

Danger of explosion!

Only use high-strength cable glands appropriate for the cable size.

Attach suitable caps to unused openings.

Caution Switch cams and limit switches are set at the factory and must not be modified.

Before connecting the mains cable, check whether the type of current, the supply voltage and the frequency stated on the type plate match the local conditions.

Caution

An incorrect supply voltage may destroy the servomotor.

To ensure electromagnetic compatibility (EMC), the signal cables must be shielded.

Caution

The screen must be connected on one side to PE (earth) or to the housing earth (earth terminal).

Lay signal and mains cables in separate cable channels.

Caution

For parallel operation: If several drives are controlled via a shared contact, every drive must be equipped with a relay for parallel operation.

Note

Lay the mains supply lines with cross-sections that conform to the relevant VDE guidelines.

Note

Use separate, shielded lines for low voltages (e.g. potentiometers, analogue signals).

Use screwed cable joints appropriate for the connection line.

Note

The stated degree of protection can only be ensured if suitable screwed cable joints are used.

- Carry out the electrical connections according to the servomotor manufacturer's instructions.

7. Commissioning

Caution

Before start-up, all components of the system must be ready for start-up.

Note

Observe the installation and operating instructions for the components used.

7.1 Checks before start-up

7.1.1 Checking the direction of rotation

The servomotor is controlled by an external controller. Therefore, the direction of rotation of the servomotor also depends on correct cabling and correct settings of the controller.

Note

Observe the installation and operating instructions for the controller and the dosing pump.

1. Switch on the dosing pump and the controller.
2. Switch the controller to manual operation.
3. Read the stroke-length value on the scale ring.
4. On the controller, increase the stroke length manually (in the "+" direction) for a few seconds.
5. Read the stroke-length value on the scale ring:
 - The stroke length has increased:
Correct direction of rotation.
 - The stroke length has decreased:
Wrong direction of rotation.
6. If the direction of rotation is wrong, check and correct cabling and controller settings. Repeat the check.
7. Switch the controller to automatic operation.

7.1.2 Adjusting the zero point

Note

Observe the installation and operating instructions for the dosing pump.

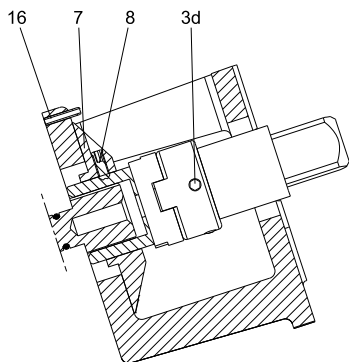


Fig. 4 Adjusting spindle with coupling

Depending on the counter-pressure of the pump, the zero point (no dosing) of the pump may be at a stroke length a little higher than 0 %. If a high precision is required, it may be necessary to adjust the zero point.

1. Switch the controller to manual operation.
2. On the controller, reduce the stroke length manually until the servomotor switches off automatically.
For versions with 4-20 mA input, set the input signal to 4 mA.
3. Loosen the grub screw (3d) in the coupling, and turn the coupling clockwise until the pump stops dosing.
4. Re-tighten the grub screw (3d) in the coupling.
5. Loosen the grub screw (8) in the scale ring (7), and turn the scale ring so that "0" covers the marker pin (16) in the motor flange.
6. Re-tighten the grub screw (8) in the scale ring.
7. Switch the controller to automatic operation.

8. Operation

Do not use the position adjustment knob at the servomotor. When using the position adjustment knob, the min./max. position can be exceeded. Exceeding the min./max. position leads to destruction of the adjusting spindle and the sliding plug in the pump.

Caution

8.1 Automatic operation

Caution

Before setting the servomotor to automatic operation, check the min./max. position, the direction of rotation and adjust the zero point.

Note

During automatic operation, all functions are controlled by an external controller.

Observe the installation and operating instructions for the controller and the dosing pump.

8.2 Manual stroke-length adjustment

In case of faults of the servomotor or in the system, it may be necessary to adjust the stroke length manually, see the servomotor manufacturer's instructions.

When the fault has been corrected, switch the system back to automatic operation.

8.2.1 Manual operation

1. Separate the servomotor from the power supply.
2. Loosen the grub screw (3d) in the coupling (see fig. 4).
3. Turn the coupling by hand until the desired stroke length is reached.

Switching back to automatic operation

1. Turn the coupling so that "0 %" covers the marker pin (16) in the motor flange.
2. Connect the servomotor to the power supply.
3. Reduce the stroke length manually until the servomotor switches off automatically.
For versions with 4-20 mA input, set the input signal to 4 mA.
For versions with potentiometer set min. stroke length.
4. Tighten the grub screw (3d) in the coupling.
5. Switch the controller to automatic operation.

8.3 Configuration of the servomotor

For the configuration of the servomotor inputs and outputs and the setting of operating modes, see the servomotor manufacturer's instructions.

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9. Maintenance



Warning

Electrical connections may only be established by qualified staff.

Observe the servomotor manufacturer's instructions supplied with the servomotor.



Warning

Tools meeting the requirements of EN 1127-1 Appendix A must be used for all operations on pumps in potentially explosive atmospheres.

There is a general ban on the use of any type of steel tool in Zone 1 if there is a risk of explosion due to materials in Explosion Group IIB and hydrogen sulphide, ethylene oxide or carbon monoxide.



Warning

Switch off the entire system before making any repairs!

Switch off the power supply before opening the cover!



Warning

Risk of explosion due to overheating of the servomotor.

Prescribed maintenance operations must be carried out at regular intervals in order to ensure continued conformity with the requirements of Directive 2014/34/EU.

9.1 Cleaning

Caution

In order to prevent heat accumulating due to dust deposits on the servomotor, the servomotor must be cleaned at regular intervals.

It is the responsibility of the operator to set suitable intervals. Dust must not be allowed to accumulate > 2 mm.

10. Disposal

This product or parts of it must be disposed of in an environmentally sound way. Use appropriate waste collection services. If this is not possible, contact the nearest Grundfos company or service workshop.

Subject to alterations.

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